

NVH Source Locator —



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NVH Source Locator is a powerful tool for identifying the source of noise and vibration (NVH) in a system. It uses a combination of sensors and software to measure the time difference of arrival (TDOA) of sound waves from different sources. This allows the user to pinpoint the exact location of the noise source, even in complex environments. The tool is designed to be easy to use and provides detailed reports on the results of the analysis.

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- [How it works](#)
- [Getting started](#)
- [Installation](#)
- [2-Sensor](#)
- [3-Sensor](#)
- [Pro+ \(3-Sen+, 4-Sensor, 4-Sen+, 3D, 3D+\)](#)
- [Materials](#)
- [FAQ](#)
- [Contact us](#)
- [Privacy policy](#)
- [Terms of use](#)
- [Pro](#)
- [Help](#)
- [About us](#)

[How it works](#)

The NVH Source Locator is a powerful tool for identifying the source of noise and vibration (NVH) in a system. It uses a combination of sensors and software to measure the time difference of arrival (TDOA) of sound waves from different sources. This allows the user to pinpoint the exact location of the noise source, even in complex environments. The tool is designed to be easy to use and provides detailed reports on the results of the analysis.

NVH

NVH Source Locator

TDOA Calculator PRO

2-Sensor

3-Sensor

3-Sen+

4-Sensor

4-Sen+

3D

3D+

Materials

Pair A-B

Pair A-C

Step 1

Calibration — Speed of Sound

Sensor spacing A-B

-

118

+

cm

External knock time delay

-

471

+

μs

Calculated speed of sound

2505 m/s

Closest: PEEK (2500 m/s)

+ Save as custom material

Step 2

Event Measurement

Event time delay

⊕ trials

-

227

+

μs

Which sensor heard it first?

Sensor A

Calculate Source Location

2-Sensor

{#the-main-tabs}

:

	1D	2D	3D
2-Sensor	1D	2D	3D
3-Sensor	2D	3D	4D
3-Sen+	3-Sensor	4-Sensor	5-Sensor
4-Sensor	2D	3D	4D
4-Sen+	3D	4D	5D
3D	3D	4D	5D
3D+	3D	4D	5D
Materials	3D	4D	5D
Help	3D	4D	5D

vs Pro: The 2-Sensor version is more expensive. However, it offers more features, such as a larger screen, more storage, and a longer warranty. The Pro version is a better value for money.

■■■■■ 2-Sensor {#2-sensor-mode}



NVH Source Locator
 TDOA Calculator PRO




2-Sensor 3-Sensor 3-Sen+ 4-Sensor 4-Sen+ 3D 3D+ Materials

Pair A-B

Pair A-C

Step 1

Calibration – Speed of Sound

Sensor spacing A-B

-

118

+

cm

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Event Measurement

Event time delay

⊕ trials

-

227

+

μs

Which sensor heard it first?

Sensor A

Calculate Source Location

2-Sensor

1:

Materials. , Mild (1020)". .

, " " .

2.

2: 2020 年 12 月 1 日

■ ■■■■■■ 2-Sensor ■■ ■■■■■■ ■■■ ■■■■■■ ■ ■■■■■■: ■■■■■■ A-B ■ ■■■■■■ A-C
(■■■■■■■ ■■ ■■■■ A-B, ■■■ ■■■■■■ ■■■■ 2 ■■■■■■).

□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ : □

3: 3D Printing and the Future of Manufacturing

- **Event-driven programming** (tEvent): **asynchronous** **non-blocking** **non-terminating**, **asynchronous** **non-blocking** **non-terminating** **asynchronous** **non-blocking** **non-terminating**
- **Event-driven programming**: **asynchronous** **non-blocking** **non-terminating** **asynchronous** **non-blocking** **non-terminating** (A **asynchronous** B)

4: **How to use the data**

A:

- $\mathbf{0} = \mathbf{0}$: $\mathbf{0}$ is the zero vector
- $\mathbf{0} = \mathbf{0}$: $\mathbf{0}$ is the zero vector
- $\mathbf{0} = \mathbf{0}$: $\mathbf{0}$ is the zero vector
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■■■■■■■■ ■ ■■■■■■■■■■ ■■■■■■■■ ■ ■■■■■■ ■■■■■■■■■■■■ (■■■ A, ■■■ B) ■ ■■■■■■■■
 ■■■■ ■■■■■■■■ ■ ■■-■■■■■.

5 (5): 11

1. 在 2019 年 12 月 31 日，公司 2019 年度财务报表已经审计，注册会计师出具了标准无保留意见的审计报告。

3-Sensor {#3-sensor-mode}

NVH

NVH Source Locator

TDOA Calculator

PRO

2-Sensor

3-Sensor

3-Sen+

4-Sensor

4-Sen+

3D

3D+

Materials

Same triangle as 3-Sensor, but calibrate & measure **all three pairs** (A-B, A-C, B-C). The solver uses all 3 TDOAs in a least-squares fit — more robust to measurement noise and anisotropic materials. The *RMS residual* tells you how consistent your measurements are.

▲ Show placement guide

Setup

Triangle side lengths

A-B side length

-

100

+

cm

A-C side length

-

90

+

cm

B-C side length

-

80

+

cm

Pair A-B

Calibration & measurement

Calibration time delay — knock outside A-B

-

400

+

μs

2,500 m/s — Closest: PEEK (2,500 m/s)

Event time delay

-

120

+

μs

⊕ trials

Which sensor heard it first?

Sensor A

3-Sensor

, —

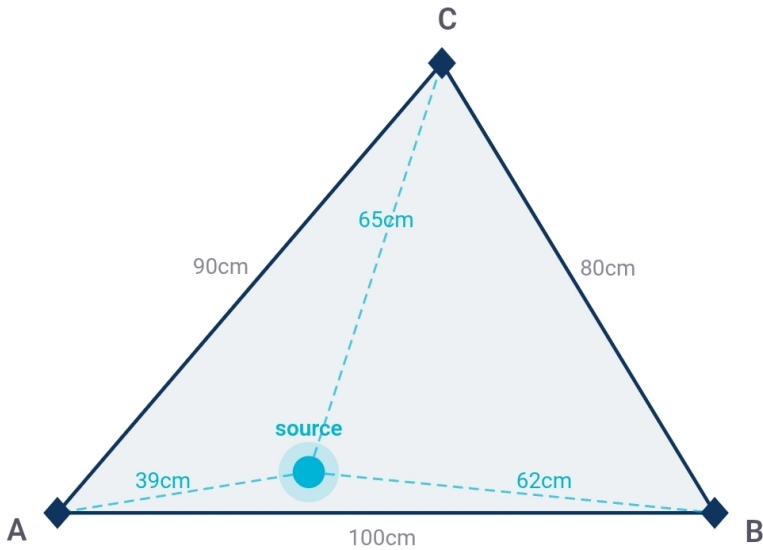
The following table shows the estimated distances between the source and the sensor corners (A-B, A-C, B-C).

The estimated distances between the source and the sensor corners (A-B, A-C, B-C) are:

- tCal: The estimated time of arrival (TOA) of the signal at the source (in seconds)
- tEvent: The estimated time of arrival (TOA) of the signal at the sensor corners (in seconds)
- Estimated distances: The estimated distances between the source and the sensor corners (in cm)

Sensor corners and source estimate

The sensor corners are labeled A, B, and C. The source estimate is labeled X, Y. The estimated distances between the source and the sensor corners are shown in the table below. The estimated distances between the sensor corners are also shown in the table below.



A, B, C = sensor corners · dot = least-squares source estimate

Copy

Share

Print result

Annotate photo

- ■■■■■■■■
- ■■■■■■■■, Mild (1020)
- ■■■■■■■■■■ ■■■■■■■■ (304)
- ■■■■■■■■ (■■■■■■)
- ■■■■■■■■
- ■■■■
- ■■■■■■■■

- **Temperature**
- **Humidity**
- **Pressure**
- **Acceleration**
- **Rotation**
- **Light**
- **Proximity**
- **Distance**

The sensor is designed to operate in a wide range of temperatures. The operating temperature range is from -40 °C to +200 °C (typical, maximum). The sensor is calibrated at 20 °C (typical, maximum).

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Operating Temperature Range

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Operating Temperature Range {#temperature-compensation}

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Operating Temperature Range

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Units & settings

×

LANGUAGE

English

DISTANCE

cm

inches

TIME

μs

ms

REFERENCE TEMPERATURE

Resets to 20 °C on next app launch

−

20

+

°C

Used for temperature-compensated material speeds. Always calibrate in-situ for best accuracy.

MEASUREMENT HISTORY

🕒 View recent calculations

⬇️ Backup

⬆️ Restore

REPORT

Report header text

e.g. EVDiag NVH Services · service@evdiag.net · +49-XXX-XXXXX

Appears at the top of every printed report.

0 / 500

DATA

🕒 Show intro screen again

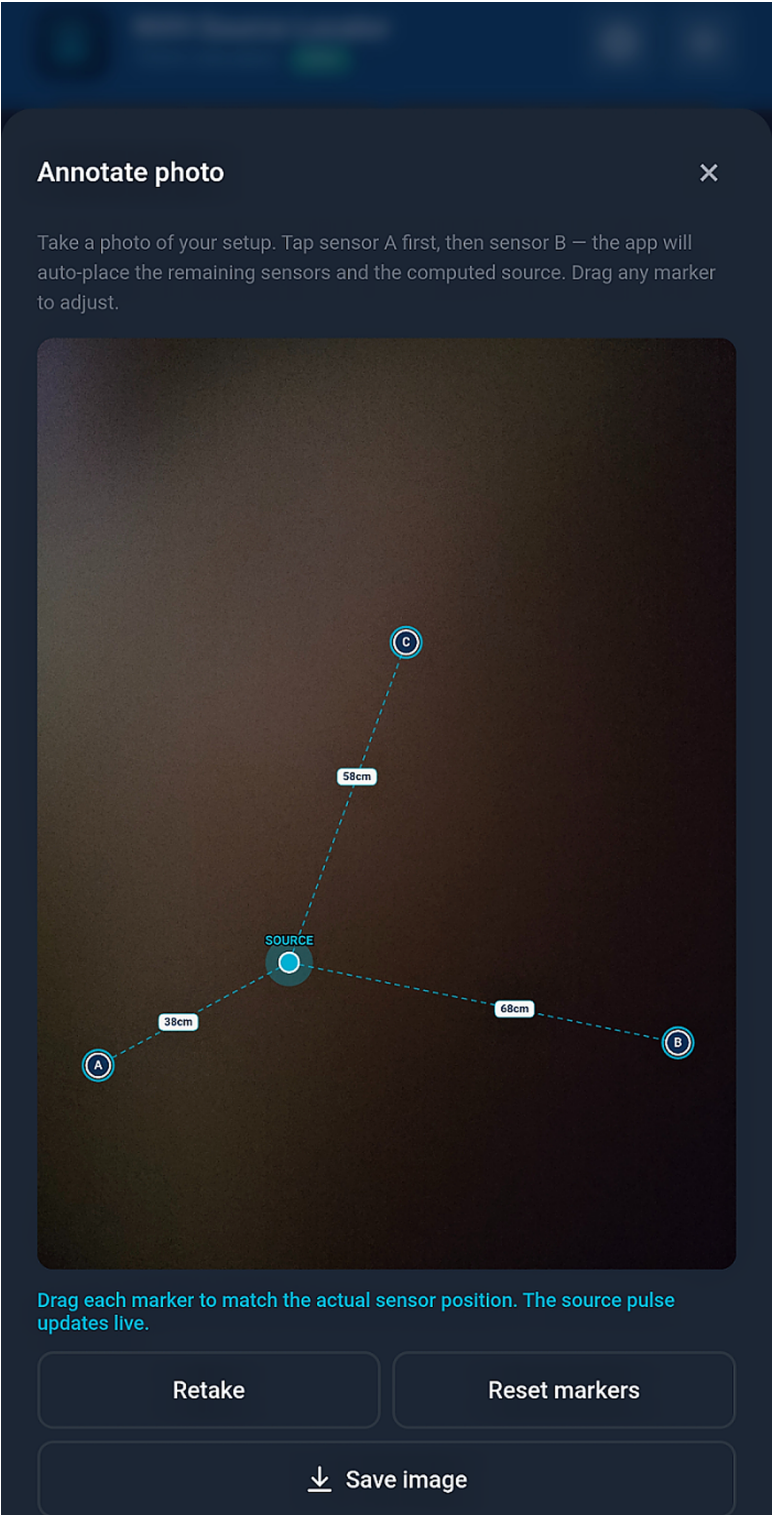
ABOUT

NVH Source Locator · TDOA Calculator for accelerometer-based source localization. Works offline. No data leaves your device.

□ □ □ □ □ □ □ □ □ □ □ □ □ □ □

■■■■■ ■■ ■■■■■■, ■■■■■■ ■■■■■■■■■■■■■■ $\neq 20\text{ }^{\circ}\text{C}$

- 
- 
-  Materials 
- Toast : „ (6 284 m/s @ 60 °C) —  N " 



■■■■■■■■■■ ■■ ■■■■■■

■■■■■■■■

- ■■■■■■■■■■ ■■■■■■■■■■ ■■■■■■■■ — ■■■■■■■■ ■■ ■■■■■■■■■■■■ ■■■■■■■■
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 - ■■■■■■■■■■■■ ■■ ■■■■■■■■■■■■ (A, B, C, D, E, F ■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■ — ■■ 6 ■■■■■■■■■■)
- ■■■■■■■■■■■■ ■■ ■■■■■■■■■■■■ ■■ ■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■ ■■■■ ■■■■■■■■ ■■ ■■■■■■■■■■

Android 打印 PDF

- Android: 使用系统自带的 PDF 阅读器，或者使用第三方 PDF 阅读器，如 Adobe Reader、Foxit Reader 等。
- iOS: 使用系统自带的 PDF 阅读器，或者使用第三方 PDF 阅读器，如 Adobe Reader、Foxit Reader 等。

打印 PDF 文件

打印 PDF 文件 → 选择要打印的 PDF 文件。在文件列表中，找到要打印的 PDF 文件，点击文件名称，即可打开文件。在文件打开后，点击右上角的“打印”按钮，即可进入打印界面。

打印 PDF 文件 {#backup-and-restore}

打印 PDF 文件 → 选择要打印的 PDF 文件。在文件列表中，找到要打印的 PDF 文件，点击文件名称，即可打开文件。在文件打开后，点击右上角的“打印”按钮，即可进入打印界面。

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打印 PDF 文件 → 选择要打印的 PDF 文件。在文件列表中，找到要打印的 PDF 文件，点击文件名称，即可打开文件。在文件打开后，点击右上角的“打印”按钮，即可进入打印界面。

打印 PDF 文件 {#settings}

打印 PDF 文件 → 选择要打印的 PDF 文件。在文件列表中，找到要打印的 PDF 文件，点击文件名称，即可打开文件。在文件打开后，点击右上角的“打印”按钮，即可进入打印界面。

Units & settings

×

LANGUAGE

English

DISTANCE

cm

inches

TIME

μs

ms

REFERENCE TEMPERATURE

Resets to 20 °C on next app launch

-

20

+

°C

Used for temperature-compensated material speeds. Always calibrate in-situ for best accuracy.

MEASUREMENT HISTORY

🕒 View recent calculations

⬇️ Backup

⬆️ Restore

REPORT

Report header text

e.g. EVDiag NVH Services · service@evdiag.net · +49-XXX-XXXXX

Appears at the top of every printed report.

0 / 500

DATA

🕒 Show intro screen again

ABOUT

NVH Source Locator · TDOA Calculator for accelerometer-based source localization. Works offline. No data leaves your device.

■■■■■■■■■■

■■■■■■■■■■	■■■■■ ■■■■■■■■■■
■■■■■■■■■■ ■■ Pro	■■■■■■■ ■■■ ■■■■■■■■■■ ■■ Pro ■■■■■■■■■■ (\$19,99)
■■■■■	■■■■■ ■■ ■■■■■■■■■■ ■■ ■■■■■■■■■■ (■■■■■■■■■■■ ■■ 30)
■■■■■	■■■■■■■, ■■■■■■ ■■■ ■■■■■■■■■■ (■■■■■■■■■■■ ■■ ■■■■■■■■■■
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- **PRO**
-
- (\$19.99;)



Works offline

Installable on Android & iOS

No data sent anywhere



Single pair A-B or A-C. Linear result bar.



Two pairs, proportional rectangle map.



Triangle mode, TDOA hyperbola solver.



49 materials. Tap to auto-fill calibration.



Mode explainers, placement guides, and a 10-item FAQ covering calibration, TDOA theory, and accuracy limits.

1

Strike the component outside the sensor pair. Enter the sensor spacing (cm) and the measured time delay (μs) – the app calculates the speed of sound in the material.

■■■■■ Help

[illegible]

- **3D printing** is a **rapid prototyping** technology
- **3D printing** is a **layer-by-layer** manufacturing process
- **3D printing** is a **digital** manufacturing process
- **3D printing** is a **flexible** manufacturing process
- **3D printing** is a **cost-effective** manufacturing process

